

Problem Statement - Hackathon

Challenge 2: AI Service Orchestrator for Informal Economy

CHALLENGE OVERVIEW

Informal service economy includes plumbers, electricians, tutors, beauticians, drivers, mechanics, AC technicians, home service providers, and small local professionals. Service discovery often happens through WhatsApp, phone calls, referrals, and informal networks, causing missed opportunities, poor matching, unpredictable pricing, weak follow-up, and limited trust. This challenge requires an agentic system that automates the end-to-end service lifecycle from natural-language request to matching, scheduling, pricing, booking, follow-up, quality feedback, and dispute handling.

PROBLEM STATEMENT

1. Understand multilingual service requests (for example Urdu, Roman Urdu, English, and mixed/code-switched language).
2. Extract service type, location, urgency, preferred time, constraints, and user preferences.
3. Discover providers using mock data, Google Maps/Places, or other APIs.
4. Rank providers using a multi-factor matching algorithm rather than distance alone.
5. Generate dynamic price quotes with transparent breakdowns.
6. Simulate booking, confirmation, reminders, service progress, feedback, reputation updates, and dispute resolution.

MANDATORY REQUIREMENT: GOOGLE ANTIGRAVITY

- Use Google Antigravity as the main orchestrator for intent understanding, matching, scheduling, pricing, booking, follow-up, and service-quality workflows.
- Show Antigravity reasoning traces for provider selection, price estimation, scheduling conflicts, confirmation actions, and dispute escalation. External LLMs, Maps/Places APIs, spreadsheets, mock datasets, and messaging APIs may be used, but Antigravity must control the agentic workflow.

SYSTEM REQUIREMENTS

- **Multilingual and noisy input handling:** Handle Urdu, Roman Urdu, English, misspellings, slang, and code-switching such as "Mujhe kal morning main AC service chahiye". Provide confidence score and confirmation questions when confidence is low.
- **Advanced provider matching:** Use at least six factors: distance/travel time, availability, rating, review recency, reliability/on-time score, skill specialization, price, capacity, cancellation rate, user preference, and risk score.

- **Skill and job complexity classification:** Classify job complexity as basic, intermediate, or complex and match to provider specialization, experience, tools, or certifications.
- **Scheduling intelligence:** Prevent double booking, include travel-time buffers, suggest alternate slots, manage waitlists, and reschedule automatically if a provider cancels.
- **Dynamic pricing:** Calculate price using demand, urgency, distance, service complexity, provider rate, loyalty discount, and surge conditions. Show breakdown and fairness to both user and provider.
- **Booking simulation:** Simulate booking confirmation, provider assignment, calendar update, SMS/WhatsApp notification, receipt, and database/spreadsheet update.
- **Service-quality loop:** Simulate en-route update, service completion checklist, photo/video evidence placeholder, customer feedback, rating adjustment, and future matching impact.
- **Dispute and escalation workflow:** Handle no-show, cancellation, quality complaint, price disagreement, overrun, refund, compensation, blacklist, or human escalation simulation.
- **Provider-side optimization:** Show provider workload balancing, fair earning opportunity, demand forecasting, or recommended time slots for provider utilization.
- **Robustness and fallback:** Handle no provider available, low-confidence language parsing, Maps/API failure, payment confirmation failure, and user preference conflicts.

EXAMPLE SCENARIO

- **User input:** "AC bilkul kaam nahi kar raha, kal subah G-13 mein technician chahiye, budget zyada nahi hai."
- **Understanding:** Service = AC repair; issue severity = high; location = G-13; time = tomorrow morning; price sensitivity = high.
- **Matching:** Ranks providers using travel time, availability, AC specialization, on-time score, review sentiment, rate, cancellation risk, and capacity.
- **Decision:** Recommends Provider A despite Provider B being closer because Provider A has higher reliability and specialized AC repair reviews.
- **Pricing:** Generates quote with visit fee, distance cost, urgency adjustment, and budget-sensitive alternative.
- **Simulation:** Books 10:00 AM slot, sends confirmation, updates booking sheet, schedules reminder, simulates provider en-route update, collects feedback, and updates reputation score.

RECOMMENDED STRESS-TEST SCENARIOS

- No suitable provider is available in the requested time window.
- A provider cancels after confirmation and the system must reschedule.
- User input is misspelled, mixed-language, or ambiguous.
- Two users request the same provider at overlapping times.
- Customer disputes price or quality after service completion. Provider has high rating but recent negative reviews and high cancellation rate.

DELIVERABLES

1. Working prototype with a mobile app as mandatory and web app as optional.
2. Demo video of 3-5 minutes showing user request -> intent extraction -> provider ranking -> pricing -> booking -> follow-up -> feedback/dispute workflow.
3. Antigravity agent trace/logs showing language parsing confidence, provider ranking rationale, scheduling decisions, price logic, action execution, and fallback behavior.
4. README with architecture, provider dataset schema, matching factors, Antigravity workflow, APIs/tools, assumptions, cost/latency analysis, baseline comparison, privacy note, and limitations.

EVALUATION CRITERIA

- Antigravity integration 20%
 - Matching and decision quality 25%
 - Multilingual robustness and edge cases 15%
 - Scheduling, pricing and service workflow 15%
 - Dispute handling, reliability and scalability 15%
 - Innovation and UX 10%
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SHARED SUBMISSION CHECKLIST

- **Working prototype:** Mobile app is mandatory for all challenges. Web app/dashboard is optional unless needed for demonstration.
- **Demo video:** 3-5 minutes for Challenges 1-3; around 3 minutes for Challenge 4. Must show agentic workflow end to end.
- **Demo video:** 2–3-minute video which shows a screen recording on how your team made use of Antigravity solutions
- **Antigravity trace/logs:** Workplan, task plan, agent observations, reasoning, decisions, tool calls, action execution, error recovery, and final outcomes.
- **README/documentation:** Architecture, data schemas, tools/APIs, Antigravity role, setup steps, assumptions, privacy note, cost/latency, scalability, baseline comparison, and limitations.
- **Baseline comparison:** Show how agentic system performs better than simple heuristic or non-agentic implementation.
- **Robustness evidence:** At least one failure, edge case, contradiction, missing data, or fallback scenario demonstrated.
- **Cost and scalability note:** Cost per operation or API call estimate; 10x/100x scaling discussion; latency or throughput estimate.

FINAL NOTES FOR PARTICIPANTS

- Do not build a simple summarizer, listing app, static dashboard, fixed-rule game, or hard-coded workflow and call it agentic.
 - Judges will look for evidence that the system observes, reasons, decides, acts, evaluates, and adapts.
 - Use realistic mock data if real APIs are unavailable. Clearly label synthetic data and avoid personal or sensitive information.
 - Prioritize a reliable end-to-end workflow over an overly broad but shallow implementation.
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KINDLY READ THROUGH THE CHALLENGES ON THE NEXT PAGES AND MAKE YOUR SELECTION AT THE END. PLEASE MAKE SURE TO READ THOROUGHLY AS ONCE A SELECTION IS MADE AND SUBMITTED, IT CANNOT BE CHANGED

To reiterate the judging criteria is as follows:

Submissions will be judged on prototype quality, Google Antigravity integration, Agentic reasoning, technical implementation, innovation, and relevance.

Judges will assess how clearly the solution demonstrates planning, decision-making, tool/API use, action simulation, and visible outcomes through the prototype, demo video, documentation, and agent traces/logs.

- **Google Antigravity Integration: 25%**
- **Agentic Reasoning & Workflow: 20%**
- **Problem Understanding & Decision Quality: 20%**
- **Action Simulation & Outcome: 15%**
- **Technical Implementation: 10%**
- **Innovation, UX & Demo Clarity: 10%**

Challenge Selection Deadline: **15th May 2026**

Challenge Submission Deadline: **20th May 2026** (*Separate Form Will be Shared*)